

Java Assignment 01: Basic Programs

Java Assignment 1 – Basic Programs

Part A – Data Types, Variables and Operators

1. Arithmetic Operations

Write a Java program to declare two integer variables and perform addition, subtraction, multiplication, division, and remainder operations. Display all the results.

2. Area of a Circle

Write a Java program to store the radius of a circle in a variable and calculate its area and circumference.

Formula:

$$\text{Area} = \pi \times r \times r$$

$$\text{Circumference} = 2 \times \pi \times r$$

3. Simple Interest

Write a Java program to accept principal amount, rate of interest and time using variables and calculate the simple interest and total amount.

Formula:

$$\text{Simple Interest} = (P \times R \times T) / 100$$

$$\text{Amount} = P + \text{Simple Interest}$$

4. Temperature Conversion

Write a Java program to convert temperature from Celsius to Fahrenheit.

Formula:

$$\text{Fahrenheit} = (\text{Celsius} \times 9 / 5) + 32$$

5. Calculate Total and Average

Write a Java program to store marks of three subjects in variables and calculate the total and average marks.

6. Even or Odd

Write a Java program to accept an integer and check whether the number is even or odd using if-else.

7. Positive, Negative or Zero

Write a Java program to accept an integer and check whether it is positive, negative, or zero using if and if-else.

8. Largest of Two Numbers

Write a Java program to accept two numbers and find the larger number using if-else. If both numbers are equal, display an appropriate message.

9. Voting Eligibility

Write a Java program to accept a person's age and check whether the person is eligible to vote.

Age $\geq 18 \rightarrow$ Eligible

Age $< 18 \rightarrow$ Not Eligible

10. Student Result

Write a Java program to accept marks of a student and check whether the student has passed or failed.

Marks $\geq 40 \rightarrow$ Pass

Marks $< 40 \rightarrow$ Fail